1.Attendance Recovery Computation

Problem Statement

Develop a Python program to calculate a student's current attendance percentage and determine the minimum number of future classes the student must attend to reach the required attendance eligibility percentage.

The program should also compare two future attendance scenarios:

- Scenario 1: The student attends every upcoming class.
- Scenario 2: The student attends 4 out of every 5 upcoming classes.

The program should display the current attendance percentage, the number of future classes required in each scenario, and whether the target is achievable.

Objectives

1. Calculate the current attendance percentage.
2. Check whether the student meets the required attendance percentage.
3. Find the minimum number of future classes needed to reach the target.
4. Compare different future attendance scenarios.
5. Display clear results to help students plan their attendance.

Assumptions

1. The student enters the total number of classes conducted and the number of classes attended.
2. The number of classes attended cannot exceed the total classes conducted.
3. The required attendance percentage must be between 0% and 100%.
4. In Scenario 1, the student attends every upcoming class.
5. In Scenario 2, the student attends four classes and misses the fifth class repeatedly.
6. The attendance percentage is calculated using

   Attendence% = (classes attended/classes conducted)*100

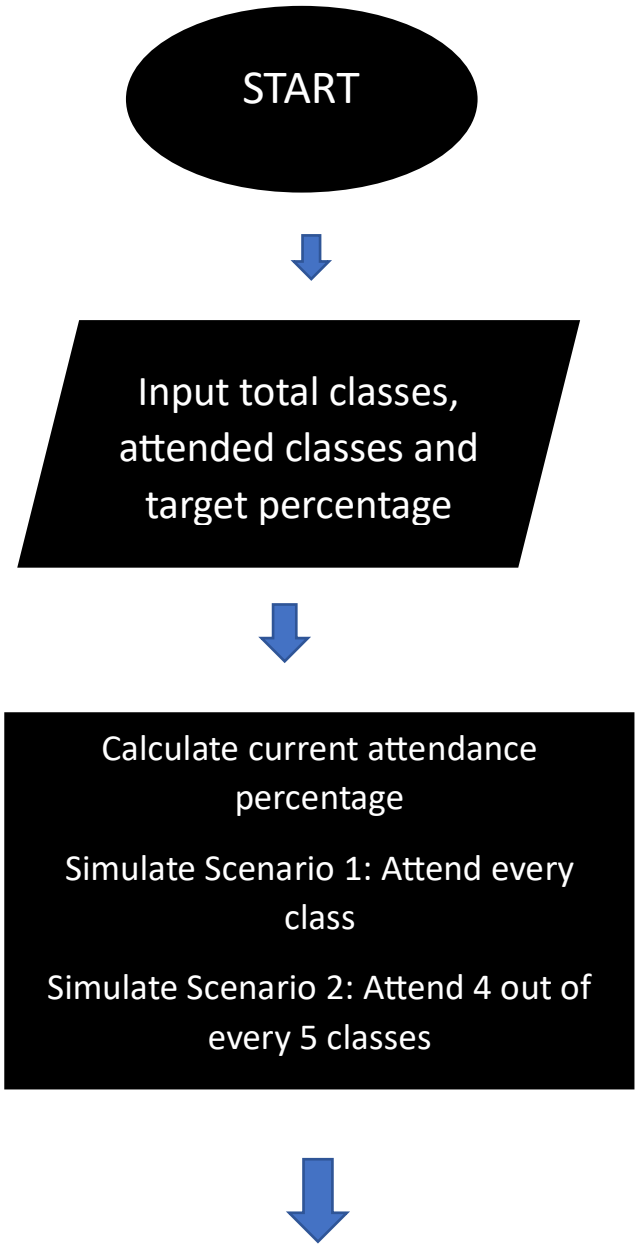
7. The program checks future classes one by one until the target is reached. If the target cannot be reached under a scenario, it reports that result.

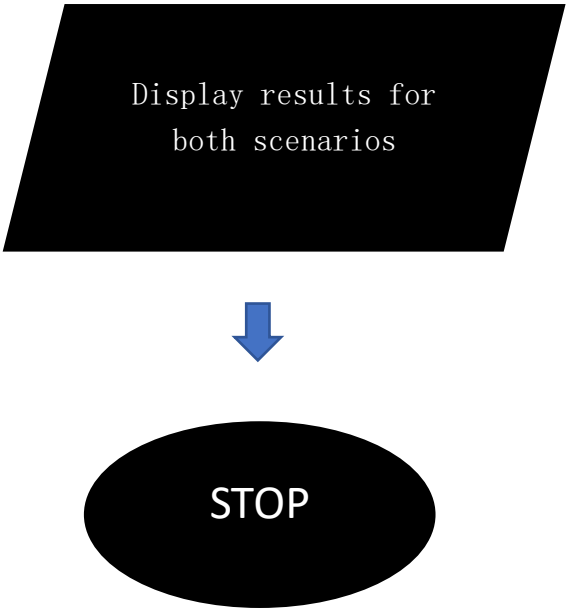
Algorithm

1. Start the program.
2. Input the total classes conducted, classes attended, and target attendance percentage.
3. Validate the input values.
4. Calculate and display the current attendance percentage.
5. If the target is already achieved, display zero additional classes required.

6. For Scenario 1, simulate attending every future class until the target is reached.
7. For Scenario 2, simulate attending four classes and missing the fifth until the target is reached.
8. If a scenario cannot reach the target, display an appropriate message.
9. Display the results of both scenarios.
10. Stop the program.

Flowchart





Pseudocode

```
START

INPUT total_classes
INPUT attended_classes
INPUT target_percentage

IF inputs are invalid
    DISPLAY error message
ELSE
    current_percentage =
        attended_classes / total_classes * 100

    DISPLAY current_percentage

    IF current_percentage >= target_percentage
        DISPLAY "Target already achieved"
    ELSE
        SIMULATE attending every future class
```

```
        FIND minimum classes needed
        DISPLAY Scenario 1 result

        SIMULATE attending 4 out of every 5 classes
        FIND minimum classes needed
        DISPLAY Scenario 2 result
    END IF
END IF

STOP
```

Python Source Code

```
total_classes = int(input("Enter total classes conducted: "))
attended_classes = int(input("Enter classes attended: "))
target = float(input("Enter target attendance percentage: "))

if (total_classes <= 0 or attended_classes < 0
        or attended_classes > total_classes
        or target < 0 or target > 100):

    print("Invalid input. Please enter valid values.")

else:
    current = attended_classes / total_classes * 100

    print(f"\nCurrent attendance: {current:.2f}%")

    if current >= target:
        print("Target already achieved.")
```

```
    print("Additional classes required: 0")

else:
    # Scenario 1: Attend every future class
    future = 0
    a1 = attended_classes
    t1 = total_classes

    if target == 100:
        scenario1 = -1
    else:
        while a1 / t1 * 100 < target and future < 100000:
            a1 += 1
            t1 += 1
            future += 1

        scenario1 = future if a1 / t1 * 100 >= target else -1

    # Scenario 2: Attend 4 out of every 5 classes
    future = 0
    a2 = attended_classes
    t2 = total_classes

    if target > 80:
        scenario2 = -1
    else:
        while a2 / t2 * 100 < target and future < 100000:
            future += 1
```

```
            t2 += 1

            if future % 5 != 0:
                a2 += 1

        scenario2 = future if a2 / t2 * 100 >= target else -1

    print("\nScenario 1: Attend every class")
    if scenario1 == -1:
        print("Target cannot be reached in this scenario.")
    else:
        print("Additional classes required:", scenario1)

    print("\nScenario 2: Attend 4 out of 5 classes")
    if scenario2 == -1:
        print("Target cannot be reached in this scenario.")
    else:
        print("Additional classes required:", scenario2)
```

Explanation of the code

`1.input()` collects the attendance details from the user.

`2.if` statements validate the inputs and check the target.

3.Arithmetic operators calculate the attendance percentage.

4.`while` loops simulate future classes until the target is reached.

5.The `%` operator identifies every fifth class in Scenario 2, which is treated as an absence.

`6.print()` displays the results.

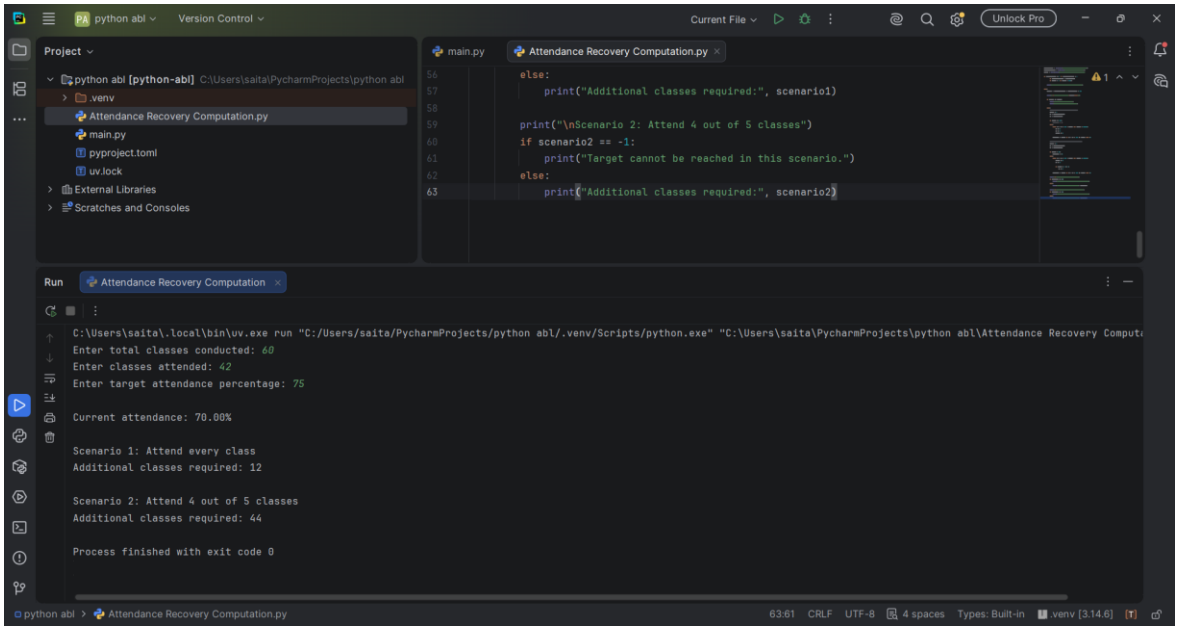
> Note:-The code uses a limit of 100,000 simulated classes to avoid an endless loop. For Scenario 1, a target of 100% cannot be reached if the student has already missed a class. For Scenario 2,

a target above 80% cannot be reached if the student is currently below the target.

Test Cases

Test	Total classes	Attended	Target	Expected result
1	60	42	75%	Scenario 1: 12 classes; Scenario 2: 60 classes
2	50	45	75%	Target already achieved; 0 classes
3	40	20	75%	Scenario 1: 40 classes; Scenario 2: 200 classes
4	60	42	90%	Scenario 1: 240 classes; Scenario 2: target cannot be reached
5	0	0	75%	Invalid input message
6	50	55	75%	Invalid input message

Output screenshot



Conclusion

The Attendance Recovery Computation program calculates a student's current attendance percentage and determines the number of additional classes required to reach the target attendance. It compares two future attendance scenarios and displays the results. Through this program, I learned how to use variables, arithmetic operators, conditional statements, loops, and input validation in Python. This program helps students understand how regular attendance improves their eligibility.